TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

AI.110

Human-Robot Interaction

COURSE OVERVIEW

Designing robots that communicate and collaborate with people: interfaces, social cues, and trust.

HUMAN-ROBOT INTERACTION

1

SESSION

60 MINUTES

ORIENTATION: ROBOTS AMONG PEOPLE

KEY TOPICS

- What is HRI
- Why human-centered design matters
- Domains: assistive, service, collaborative
- Course roadmap

TEACHING FOCUS

Put the human at the center of robot design.

KEY TAKEAWAY A robot's user is usually a person.

2

SESSION

60 MINUTES

COMMUNICATION MODALITIES

KEY TOPICS

- Speech and natural language
- Gestures and body language
- Facial expression and gaze
- Haptics and touch

TEACHING FOCUS

Give robots multiple ways to express intent.

▲ **KEY TAKEAWAY** Communication is multimodal, not just words.

3

SESSION

60 MINUTES

SOCIAL CUES & PROXEMICS

KEY TOPICS

- Personal space and distance
- Turn-taking and attention
- Joint attention
- Social norms for robots

TEACHING FOCUS

Make robots behave socially appropriately.

KEY TAKEAWAY Respecting space builds trust.

4

SESSION

60 MINUTES

TRUST & ACCEPTANCE

KEY TOPICS

- Factors influencing trust
- Anthropomorphism and uncanny valley
- Transparency and explainability
- User studies and evaluation

TEACHING FOCUS

Design robots people will trust and use.

KEY TAKEAWAY Trust is earned through predictable behavior.

5

SESSION

60 MINUTES

COLLABORATIVE & ASSISTIVE ROBOTS

KEY TOPICS

- Cobots in industry
- Assistive robots for disabilities
- Teleoperation and shared control
- Safety in close interaction

TEACHING FOCUS

Build robots that work beside people.

KEY TAKEAWAY Collaboration requires shared understanding.

HUMAN-ROBOT INTERACTION

6

SESSION

60 MINUTES

HRI ETHICS & REVIEW

KEY TOPICS

- Privacy and surveillance
- Dependency and manipulation
- Course recap and assessment
- Toward social robotics (AI.312)

TEACHING FOCUS

Interact ethically with vulnerable users.

KEY TAKEAWAY Respecting persons is the foundation of HRI.